

However, at present the labour costs involved in aerial delivery are very high. In most countries people are allowed to operate and monitor only one drone at a time. They also need to monitor the airspace in which the drone operates. This increases the labour cost involved, with each drone requiring an operator. Unless regulations allow a single operator to manage at least 20 drones, the cost of operating drones cannot even compete with other delivery options.

At present, drones are highly regulated, and many areas are out of bounds owing to security and privacy concerns. Nevertheless, some of the industry majors are finding drone delivery to be an attractive option. Companies like Skyports and Airbus have demonstrated shore-to-ship deliveries using drones. Mattnet, another commercial drone service provider in Europe, recently launched one of the longest urban drone delivery routes to transport diagnostic samples between hospitals and laboratories in Zurich, Switzerland.

Walmart and Amazon are already delivering through drones in parts of Arizona, Arkansas, California, Florida, North Carolina, Texas, Utah, and Virginia. Walmart has 37 stores set up for drone delivery to homes and businesses. It has partnered with drone delivery service DroneUp Delivery to deliver packages that weigh up to 4.5kg. The service functions only during daytime and, according to a Walmart statement, they have more than 10,000 items available for drone delivery; the orders are delivered in around half an hour.

Amazon has started drone deliveries in parts of Northern California and Texas using its in-house Amazon Prime Air fleet of drones, which can carry items up to 2.25kg. The deliveries are monitored by Amazon employees and dropped in



Famous supermarket chain Carrefour has partnered with Goggo Network to trial autonomous delivery using a multi-stop capable AV (Courtesy: Carrefour)

customers' backyards. This service too is available only in the daytime.

Sky delivery start-up Flyrtex promises five-minute delivery of food from local restaurants to the customer's backyard. They too operate in parts of North Carolina and Texas in the USA.

In a January 2023 report on drones, McKinsey researchers observed that, "For drones to become truly cost-competitive across the board, operators will need to be able to shift their focus from observing airspace to operating drones, and the number of drones per operator will need to increase significantly. That, in turn, will require technology and regulation to advance sufficiently for a single operator to manage as many as 20 drones in a densely used airspace. These advances would include autonomous drone flight in which drones fly with limited human intervention, unmanned traffic management systems, and sense-and-avoid solutions. Once those innovations are in place, regulations will need to evolve, enabling larger numbers of drones per operator. As those shifts begin to happen, the potential cost advantage of drone delivery will begin to grow."

Self-driving trucks or vans

Self-driving trucks or vans from companies like Volvo, Caterpillar, and Komatsu have long been used off-roads for hauling goods in industries like mining, agriculture, and construction. They have also been used on roads

in private environments, say, to transport goods from parts warehouses to manufacturing units within a private campus.

As the next step, now most major truck makers as well as start-ups like Aurora, Embark, Kodiak, Plus, TuSimple, Waabi, and Waymo are working on Level-4 autonomy on highways or open roads. Many claim that their technologies will be feature-ready this year. Implementing

autonomous long-haul deliveries is every truck maker's dream. As driving on highways is a monotonous task the drivers would be happy to be relieved of it, unlike short, local trips.

In a demonstration by Kodiak Robotics and U.S. Express, last March, a self-driving eighteen-wheeler travelled non-stop for five days, carrying goods between Dallas and Atlanta. It did four round trips, which means more than 10,000km, in five days. This would have taken human drivers ten days, with regular breaks in between. The trip was, of course, supervised by drivers taking turns behind the wheel. And according to reports by the company, they had to intervene more than once.

Although highways are more structured and easier to manoeuvre than city roads, there is still a lot for trucks to learn, including changing lanes at the right time and knowing when to stop. Hence, even the so-called fully-autonomous trucks will continue to have drivers in them in the near future.

Another class of autonomous trucks is for middle- and last-mile delivery, that is, for transporting goods between warehouses or to local stores. This requires travelling on public roads, which is quite a challenge.

Microsoft is planning to invest more than \$10 million in Gatik, a California based company that is actively championing the use of autonomous trucks on public roads. Its customers include Walmart, Georgia-Pacific, KBX, and Pitney-Bowes. They use medium-